Explaining Legal Equation Transformations

Answer Key

Algebra 1

Quick Answers

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1. 5, 22
2. 32
3. $2x + 10$
4. 3, 12
5. -4, ∞, ¡built-in function open¿
6. 13
7. 0, 4
8. 4
9. $-\infty$, -1, *loopen*
10. 7

Common wrong answers

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1. *If they got 8: undid the multiplication before the addition, dividing only the 22 by 3*
4. *If they got 6: subtracted 3 from the right side instead of adding 3 to both sides, which gives 9/5*
8. *If they got 5: substituted into the changed equation instead of the original one*

How to use this key. A step is legal when the new equation has the same solution set as the old one. Credit the *reason*, not just the arithmetic: a student who reaches the right number while writing "subtracted 7" on a line where 7 was subtracted from one side only has not yet learned this skill.

1. Solve $3x + 7 = 22$ with reasons, then check.
Solution: From $3x + 7 = 22$, subtract 7 from *both* sides — Subtraction Property of Equality — to get $3x = 15$. Then divide both sides by 3, which is legal because 3 is not zero — Division Property of Equality. That gives $\boxed{x = 5}$.
Check: substituting into the original left side, $3(5) + 7 = 15 + 7 = \boxed{22}$, which matches the right side, so the solution set really was preserved. A student who divides first, turning 22 into $\frac{22}{3}$ and then subtracting 7, lands at $x = \frac{22}{3} - 7$; putting that back gives 8, not 22, which is exactly how the check catches an illegal order.

2. Solve $\dfrac{x}{4} - 3 = 5$ with reasons.
Solution: Add 3 to both sides — Addition Property of Equality — giving $\dfrac{x}{4} = 8$. Multiply both sides by 4 — Multiplication Property of Equality — giving $\boxed{x = 32}$. The addition comes first because the subtraction is the operation applied *last* to x on the left, and legal steps undo operations in reverse order.

3. Rewrite $2(x+5)$ and name the property.
Solution: The Distributive Property gives $2(x+5) = 2 \cdot x + 2 \cdot 5$, that is $\boxed{2x+10}$. This one is different in kind from the others: nothing was done to *both* sides. Only one side was rewritten into a form equal to it for every value of x, so no equation containing it can gain or lose a solution. That is the second family of legal transformations — rewriting a side using an identity.

Kind of step	**Why it is legal**
Same operation on both sides	Both sides stay equal, so solutions are kept
Rewrite one side by an identity	That side's value is unchanged for every x

4. Devi: $5x - 3 = 12 \to 5x = 9 \to x = \frac{9}{5}$.
Solution: The second line is the illegal one. Devi subtracted 3 from the right side only ($12 - 3 = 9$) instead of adding 3 to both sides. Nothing was done to the left side at all, so the two sides were no longer changed equally and the solution set moved.
Legal version: add 3 to both sides — Addition Property of Equality — giving $5x = 15$; then divide both sides by 5 — Division Property of Equality — giving $\boxed{x = 3}$.
Check: $5(3) - 3 = 15 - 3 = \boxed{12}$, matching the right side. Devi's value fails the same check: $5 \cdot \frac{9}{5} - 3 = 6$, not 12.

5. Solve $-2x < 8$.
Solution: Divide both sides by -2. Dividing an inequality by a negative number reverses the direction of the symbol, because multiplying by a negative reflects every number across zero and reverses their order. So $-2x < 8$ becomes $\boxed{x > -4}$. Testing a value confirms it: $x = 0$ gives $0 < 8$, true, and 0 is in the solution set; $x = -10$ gives $20 < 8$, false, and -10 is not.

6. Solve $4(x-2) = 3x + 5$ with reasons.
Solution: Distributive Property on the left: $4x - 8 = 3x + 5$. Subtract $3x$ from both sides — Subtraction Property of Equality (a variable term is a quantity like any other): $x - 8 = 5$. Add 8 to both sides — Addition Property of Equality: $\boxed{x = 13}$. Every line kept the same solution, so the last line's solution is the first line's solution.

7. $x^2 = 4x$, divided by x.
Solution: The Division Property of Equality requires the divisor to be *nonzero*, and here the divisor is x itself, which may be zero. Dividing by it silently assumes $x \neq 0$ and throws that possibility away, so the step is not legal as written; it loses a solution rather than gaining a false one.
Legal route: subtract $4x$ from both sides to get $x^2 - 4x = 0$, factor to $x(x-4) = 0$, and apply the zero-product property: $\boxed{x = 0 \text{ or } x = 4}$. The discarded root really was a solution, since $0^2 = 4 \cdot 0$.

8. Ravi: $2x + 6 = 10 \to x + 6 = 5$.
Solution: Dividing both sides by 2 would be legal, but Ravi divided only *some* terms: the $2x$ and the 10 were halved while the 6 was left alone. A division property applies to each side as a whole, not term by term.
Ravi's equation solves to $x = -1$. Substituting that into the original left side gives $2(-1) + 6 = \boxed{4}$, and the original right side is 10, so $x = -1$ is not a solution of the original equation. The two equations are therefore not equivalent. Substituting into Ravi's own equation instead would give 5 and hide the error — always test a suspect step against the *original*.

9. Solve $-3(x - 2) \geq 9$.
Solution: Distributive Property: $-3x + 6 \geq 9$. Subtract 6 from both sides — Subtraction Property of Equality: $-3x \geq 3$. Divide both sides by -3; this is the line where the symbol turns around, because dividing by a negative reverses order. Result: $\boxed{x \leq -1}$. Check with $x = -5$: $-3(-5 - 2) = 21 \geq 9$, true.

10. One illegal line in the solution of $2(x + 3) = 4x - 8$.
Solution: The illegal line is the second, $2x+3 = 4x-8$. It should have used the Distributive Property, which multiplies the 2 by *both* terms inside the parentheses: $2(x + 3) = 2x + 6$, not $2x + 3$. The third line is then correct arithmetic applied to a wrong equation, which is why naming the first bad line matters.
Correct solution: $2x + 6 = 4x - 8$; subtract $2x$ from both sides: $6 = 2x - 8$; add 8 to both sides: $14 = 2x$; divide both sides by 2: $\boxed{x = 7}$.